

COVID-19 NHS Pathways Calls, England, 2020

Overview

This dataset contains aggregated NHS Pathways call data from England during the COVID-19 pandemic in 2020. NHS Pathways is the clinical triage system used by NHS 111 and other urgent care services. The data captures calls related to potential COVID-19 symptoms, broken down by date, sex, age group, and clinical commissioning group (CCG). Note that these are aggregated call counts, not individual case records.

Source: NHS digital dashboards; cleaned by Quentin Leclerc.

Loading the data

```
using DataFrames
using Outbreaks
df = Outbreaks.covid19_england_nhscalls_2020()
size(df)
```

(253670, 11)

First few rows

```
first(df, 6)
```

	site_type	date	sex	age	ccg_code	ccg_name	
	String15	Date	String7	String7	String15	String	
1	111	2020-03-18	female	missing	e38000062	nhs_gloucestershire_ccg	...
2	111	2020-03-18	female	missing	e38000163	nhs_south_tyneside_ccg	...
3	111	2020-03-18	female	0-18	e38000001	nhs_airedale_wharfedale_and_craven_ccg	...
4	111	2020-03-18	female	0-18	e38000002	nhs_ashford_ccg	...
5	111	2020-03-18	female	0-18	e38000004	nhs_barking_and_dagenham_ccg	...
6	111	2020-03-18	female	0-18	e38000005	nhs_barnet_ccg	...

Summary statistics

```
describe(df)
```

	variable	mean	min	median	max	
	Symbol	Union...	Any	Any	Any	
1	site_type		111		999	...
2	date		2020-03-18	2020-05-30	2020-09-20	...
3	sex		female		unknown	...
4	age		0-18		missing	...
5	ccg_code		e38000001		zc040	...
6	ccg_name		nhs_airedale_wharfedale_and_craven_ccg		null	...
7	count	16.1684	1	4.0	1704	...
8	postcode		NA		yo16ga	...
9	nhs_region		East of England		South West	...
10	day	81.3931	0	73.0	186	...
11	weekday		monday		weekend	...

Daily call volume

```
using StatsBase
daily = combine(groupby(df, :date), :count => sum => :total_calls)
sort!(daily, :date)
first(daily, 10)
```

	date	total_calls
	Date	Int64
1	2020-03-18	128429
2	2020-03-19	152090
3	2020-03-20	137217
4	2020-03-21	122668
5	2020-03-22	137143
6	2020-03-23	165332
7	2020-03-24	139417
8	2020-03-25	130453
9	2020-03-26	112282
10	2020-03-27	114668

```
println("Date range: $(minimum(daily.date)) to $(maximum(daily.date))")
println("Total calls: $(sum(daily.total_calls))")
println("Peak day: $(daily.date[argmax(daily.total_calls)]) with $(maximum(daily.total_calls)) calls")
```

Date range: 2020-03-18 to 2020-09-20
Total calls: 4101446
Peak day: 2020-03-23 with 165332 calls